Your position-keeping defence, tested on 357 decisions

D1 against the first pit stop of every audited Formula 1 car, 2022–2026

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Why I tested it

An exact optimiser I built stops later than Formula 1 teams do on 80% of decisions, by a median of 12 laps. I proposed two explanations, tested both, and lost both. Yours is the third, and the only mechanism I have found that predicts stopping before the time optimum — the right direction. So I implemented D1: efficient while $\theta \cdot l_u + \varepsilon < g$, expiring at a tyre age of $(g - \varepsilon)/\theta$, with the earliest undefendable undercut the lap after.

The result is negative

Across 254 decisions carrying a prediction, D1's earliest undefendable undercut lands a median of +55 laps from the real stop, against +9 for my single-car optimiser. It is closer on 41 and further on 207. On 57% the predicted lap is past the end of the race.

Fed with the gaps and slopes I measure, the condition says a leader can defend an undercut for longer than the race lasts, so a chaser should never undercut. Teams undercut at a median lap 19.

Why, as a number rather than a story

Inverting the condition at the tyre age of the stop teams actually made gives the degradation rate at which D1 would have expired exactly then: $\theta^* = (g - \varepsilon) / l_u(\text{real})$.

Over 233 decisions with a resolved slope the median θ^* is 0.185 s/lap against a measured 0.045 — a ratio of 3.4×, quartiles 1.4 to 8.6, larger on 79% of them. A rate of 0.185 s/lap sits at the 99th percentile of every slope I fit in five seasons.

So this is not a calibration gap. For your equilibrium to reproduce what teams do, tyres would have to degrade at a rate the data almost never shows.

My reading, which I have not established

$\theta \cdot l_u$ prices the defender's loss over a single lap on a tyre l_u laps old. At my measured median of 0.045 s/lap and a tyre the age teams actually stop at, that is under a second, which the median 3.4 s lead covers easily. What wins a real undercut is the advantage of a tyre on its first flying lap over one at the end of its life, and a linear slope has no term for that. Measuring it directly is a different estimator and I have not attempted it.

Two measurements that hold regardless

91 of the 357 decisions belong to a car with nobody ahead within one lap. The race leader has no undercut to perform, so a quarter of the set carries no prediction under any two-car model. For the other 266, the median gap to the leader is 3.4 s, and on 260 of 266 it is smaller than the circuit's measured pit loss. Reachability is not the binding constraint — the opposite of what I expected to find.

Three defects I found while building this

All three inflated the D1 window, so all three made this negative result look stronger than it is. Comparing cumulative race time with no bound found rivals a full lap away; the nearest car ahead is now capped at one green lap and the widest retained gap is 36 s. The expiry is a tyre age and I was comparing it against a lap number, which agree on only 78% of these decisions. And my pit-loss table names circuits differently from my lap files, which still drops 85 of 266 values if joined on the slug. Each now has a test, as does the conclusion itself.